

IEOR 4404: Simulation

Lecture 6: Composition Method

Composition Method

- Generate random variables with “layered” structure, or “mixtures”
- Discrete mixtures:

$$P(X \leq x) = \sum_{j=1}^m P(Y_j \leq x)P(J = j)$$

or

$$P(X = x) = \sum_{j=1}^m P(Y_j = x)P(J = j)$$

- X is a J -mixture of $Y_1, \dots, Y_m$
- Law of total probability: Conditional on $J = j$, distribution of X follows that of Y_j

Composition Method

To generate X ,

- Generate J (using ITM for example)
- Given $J = j$, generate Y_j

The output follows the correct distribution

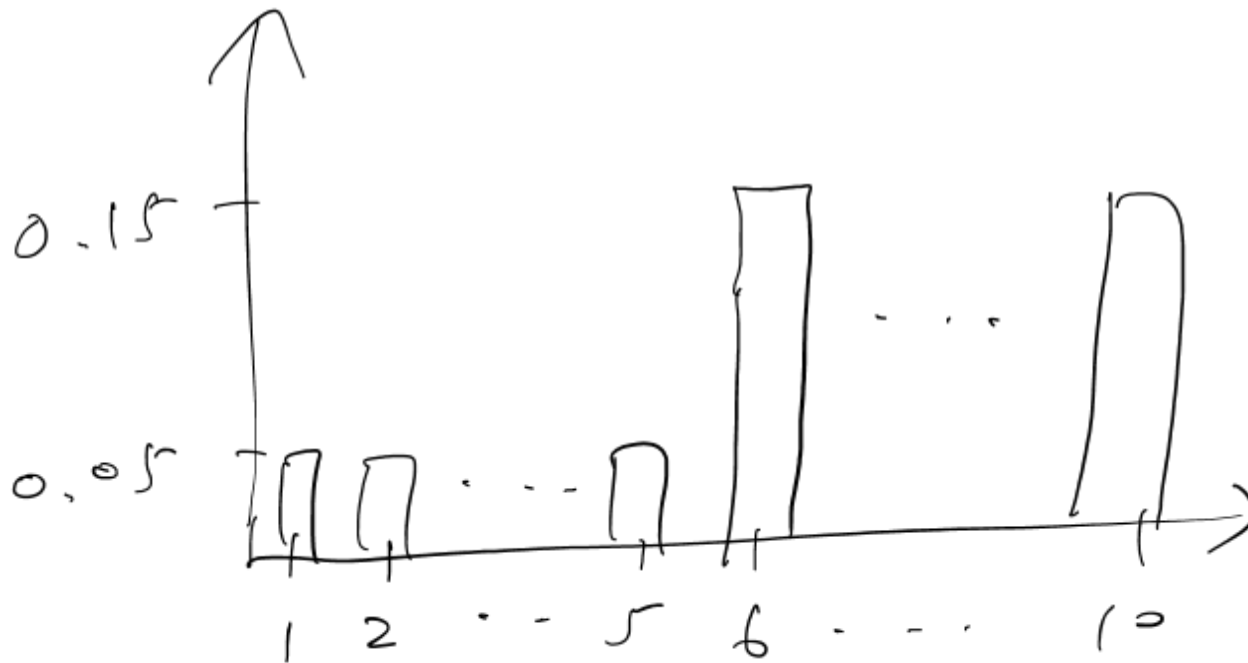
$$\begin{aligned} & P(\text{output} \leq x) \\ &= \sum_{j=1}^m P(J = j) P(\text{output} \leq x \mid J = j) \\ &= \sum_{j=1}^m P(J = j) P(Y_j \leq x) \end{aligned}$$

Example

X is a discrete r.v. with pmf

$$P(X = x) = \begin{cases} 0.05, & x = 1, 2, \dots, 5 \\ 0.15, & x = 6, 7, \dots, 10 \end{cases}$$

Generate X



Define

$$J = \begin{cases} 1 & \text{if } X = 1, \dots, 5 \\ 2 & \text{if } X = 6, \dots, 10 \end{cases}$$

"smaller"

"bigger"

$$P(J=1) = 0.25$$

$$P(J=2) = 0.75$$

$$Y_1 \sim \text{Uniform on } 1, \dots, 5$$

$$Y_2 \sim \text{Uniform on } 6, \dots, 10$$

Composition method:

- Generate J (using $1\tau M$)

- If $J = 1$, generate γ_1 (using $1\tau M$)

else if $J = 2$, generate γ_2 (using $1\tau M$)

To check the output is correct,

$$P(\text{output} = 1)$$

$$= P(J = 1) P(Y_1 = 1)$$

$$= 0.25 \times \frac{1}{5}$$

$$= 0.05$$

$$P(\text{output} = 2) = \dots = 0.05, \dots$$

Example

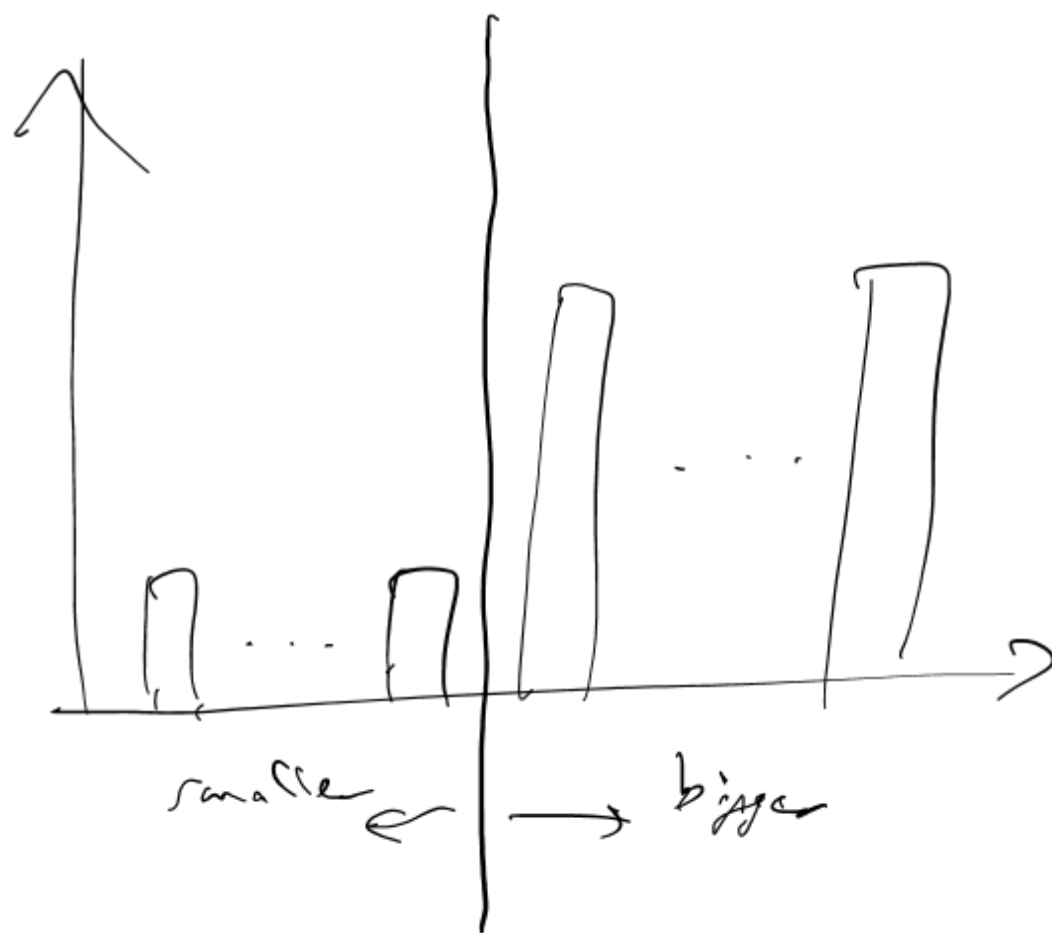
Composition method:

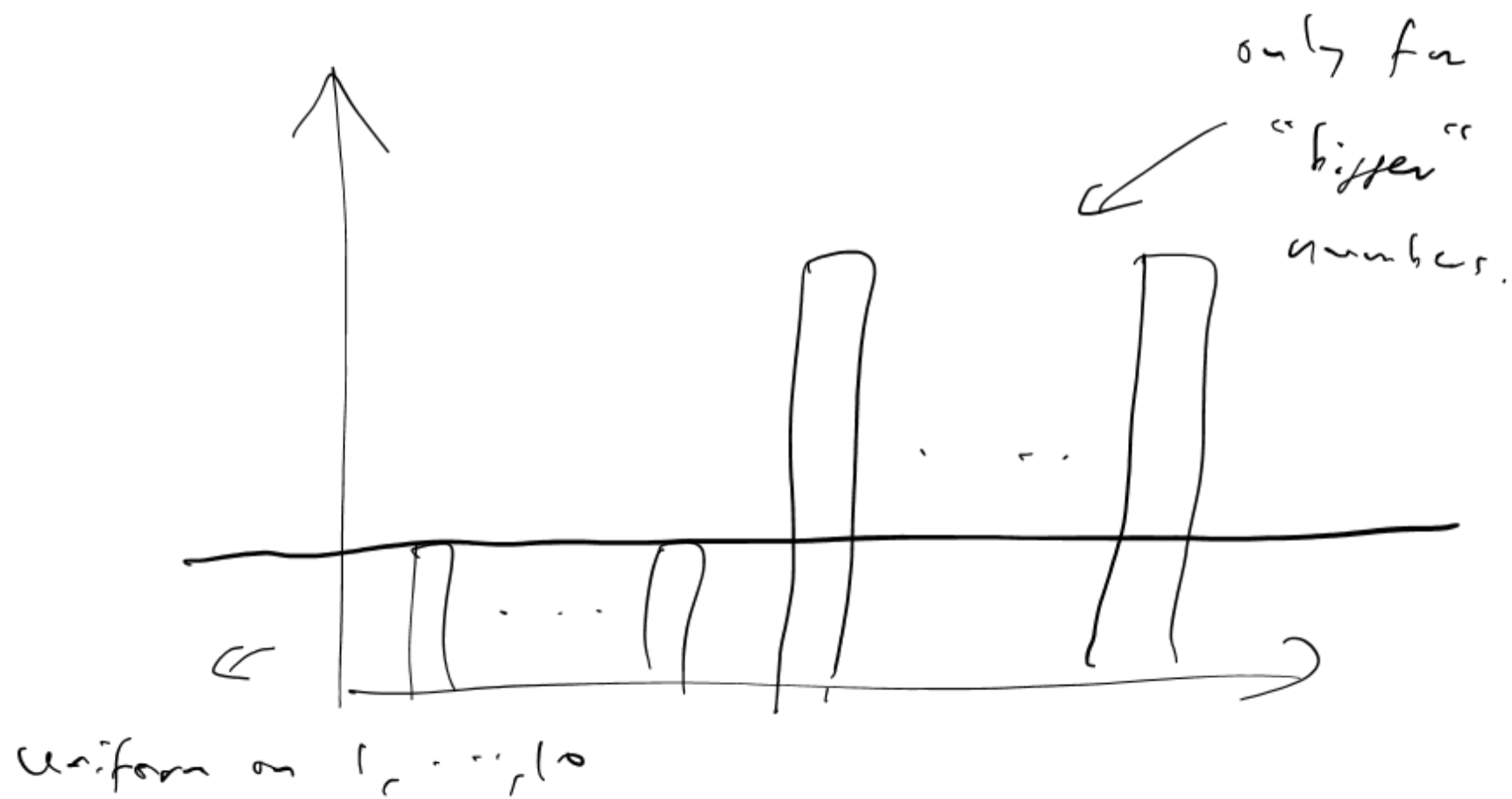
- Generate J where $P(J = 1) = 0.25, P(J = 2) = 0.75$
- If $J = 1$, generate Y_1 uniform over $\{1, 2, \dots, 5\}$; if $J = 2$, generate Y_2 uniform over $\{6, 7, \dots, 10\}$

Pseudo code:

- Generate $U \sim \text{Unif}(0,1)$
- Generate $V \sim \text{Unif}(0,1)$
- If $U \leq 0.25$, set $X = \underbrace{[5V] + 1}_{\text{uniform on } 1, \dots, 5}$; else set $X = \underbrace{[5V] + 6}_{\text{uniform on } 6, \dots, 10}$

Is this the only way to use the composition method in this example?





Example

We can also let:

- $P(J = 1) = 0.5, P(J = 2) = 0.5$
- If $J = 1$, generate Y_1 uniform over $\{1, 2, \dots, 10\}$; if $J = 2$, generate Y_2 uniform over $\{6, 7, \dots, 10\}$

We can check that the distribution of the output is still correct

To check this generates correct output,

$$\begin{aligned}P(\text{output} = 1) &= P(J=1) P(Y_1 = 1) \\&= 0.5 \times \frac{1}{10} \\&= 0.05\end{aligned}$$

$$P(\text{output} = 2) = \dots = 0.05$$

✓
✓
✓

$$P(\text{output} = 6) = P(J=1) P(Y_1=6) \\ + P(J=2) P(Y_2=6)$$

$$= 0.5 \times \frac{1}{10} + 0.5 \times \frac{1}{5}$$

$$= 0.05 + 0.1$$

$$= 0.15$$

∴

Example

Composition method:

- Generate J where $P(J = 1) = 0.5, P(J = 2) = 0.5$
- If $J = 1$, generate Y_1 uniform over $\{1, 2, \dots, 10\}$; if $J = 2$, generate Y_2 uniform over $\{6, 7, \dots, 10\}$

Pseudo code:

- Generate $U \sim Unif(0,1)$
- Generate $V \sim Unif(0,1)$
- If $U \leq 0.5$, set $X = \lfloor 10V \rfloor + 1$; else set $X = \lfloor 5V \rfloor + 6$

$\underbrace{\lfloor 10V \rfloor + 1}_{\text{unif on } 1, \dots, 10} \quad \underbrace{\lfloor 5V \rfloor + 6}_{\text{unif on } 6, \dots, 10}$

Composition Method

- In the continuous case, we consider a mixture of a “continuum” set of random variables
- Continuous mixture:

$$P(X \leq x) = \int_0^{\infty} P(Y_t \leq x) f_T(t) dt$$

or density

$$f(x) = \int_0^{\infty} f_{Y_t}(x) f_T(t) dt$$

- X is a T -mixture of $\{Y_t: t \geq 0\}$

Composition Method

To generate X ,

- Generate T
- Given $T = t$, generate Y_t

The output follows the correct distribution

Example

Generate the two-sided exponential variable with density

$$f(x) = \frac{1}{2} e^{-|x|}, -\infty < x < \infty$$

Define

$$J = \begin{cases} 1 & \text{if } X > 0 \\ 2 & \text{if } X < 0 \end{cases}$$

"positive"

"negative"

$$Y_1 \sim \text{Exp}(1)$$

$$Y_2 \sim -\text{Exp}(1)$$

Composition method:

- Generate J
- If $J=1$, generate $Y_1 \sim \text{Exp}(1)$, set $X=Y_1$
- else if $J=2$, generate $Y_2 \sim \text{Exp}(1)$
set $X=Y_2$.

To check correctness,

$$\begin{aligned} P(X \leq \text{output} \leq X+\Delta x) &= P(J=1) P(X \leq Y_1 \leq X+\Delta x) \\ &= \frac{1}{2} e^{-X} (\Delta x) \quad \text{for } X > 0 \end{aligned}$$

and

$$P(x \leq \text{output} \leq x + \delta x) = P(J=2) P(x \leq Y_2 \leq x + \delta x)$$

$$= \frac{1}{2} e^x \delta x \quad \text{for } x < 0$$

$$\Rightarrow P(x \leq \text{output} \leq x + \delta x) = \frac{1}{2} e^{-|x|} \delta x$$

for $-\infty < x < \infty$

Cautions

In the composition method, it is not necessary that the mixing variable and individual variables conditional on mixing are both discrete or continuous. Instead, it could be that one of them is discrete and another one continuous, or vice versa.

Sometimes the composition structure arises naturally from the problem, but sometimes not.

When the composition structure can be identified, the method can be faster than other more direct approaches, such as ITM and A-R.

Example 2

A production facility produces a Poisson number of products with mean t , in the time interval $[0, t]$. An inspector comes at a random time $T \sim \text{Exp}(1)$. Let X be the number of products that the inspector sees in $[0, T]$. Generate X using the composition method.

Mixing variable

$$T \sim \text{Exp}(1)$$

Given $T=t$,

$$Y_t \sim \text{Poi}(t)$$

To generate X ,

- generate $T \sim \text{Exp}(1)$

- Given $T=t$, generate $Y_t \sim \text{Poi}(t)$

Set $X = Y_t$.

Example 3

Generate X with density

$$f(x) = \int_0^{\infty} 2e^{-\frac{x}{t}-t^2} dt, \text{ for } x > 0.$$

using the composition method

Can use cont. mixing variable T to produce a cont. output at the end.

$$f(x) = \int_0^{\infty} \underbrace{f_{Y_t}(x)}_{\text{pdf of } Y_t} \cdot \underbrace{f_T(t)}_{\text{pdf of } T} dt$$

$$f_{Y_t}(x) = e^{-\frac{x}{t}} \quad \text{for } x > 0$$

$$f_T(t) = 2e^{-t^2} \quad \text{for } t > 0$$

This is not valid because

$f_{Y_t}(x)$ is not a pdf

$$f_{Y_t}(x) = \frac{1}{t} e^{-\frac{x}{t}} \quad \text{for } x > 0$$

$$f_T(t) = 2t e^{-t^2} \quad \text{for } t > 0$$

Here $f_{Y_t}(x)$ is a valid pdf.

$$Y_t \sim \text{Exp}\left(\frac{1}{t}\right)$$

Note: If $f_{Y_t}(x)$ is a valid pdf, then $f_T(t)$ is automatically also a valid pdf.

Composition method:

• Generate $T \sim f_T(t) = 2te^{-t^2}, t > 0$

• Given $T = t$, generate $Y_t \sim \text{Exp}(\frac{1}{t})$

Set $X = Y_t$.

1 TM for generating T :

$$\text{cdf: } \int_0^t 2se^{-s^2} ds = -e^{-s^2} \Big|_0^t$$

$$= 1 - e^{-t^2}$$

$$\text{Set } u = 1 - e^{-t^2}$$

$$\Rightarrow t = \sqrt{-\ln(1-u)}$$

17M: • Generate $U \sim \text{Unif}(0, 1)$
• Output $\sqrt{-\log(1-U)}$